AI.411

Robot Design

COURSE OVERVIEW

Designing complete robots: requirements, concept design, prototyping, and integration.

ROBOT DESIGN

1

SESSION

60 MINUTES

ORIENTATION: DESIGNING ROBOTS

KEY TOPICS

- The design process
- Requirements and constraints
- Concept generation
- Robot design disciplines

TEACHING FOCUS

Create robots from first principles.

KEY TAKEAWAY Design begins with purpose.

ROBOT DESIGN

2

SESSION

60 MINUTES

MECHANICAL DESIGN

KEY TOPICS

- Kinematic design
- Structure and materials
- Actuator integration
- CAD and prototyping

TEACHING FOCUS

Design the robot's body.

KEY TAKEAWAY The body enables the behavior.

ROBOT DESIGN

3

SESSION

60 MINUTES

ELECTRICAL & ELECTRONICS

KEY TOPICS

- Power systems
- Wiring and sensors
- Microcontroller selection
- PCB design

TEACHING FOCUS

Power and wire the machine.

▲ **KEY TAKEAWAY** Electronics are the robot's nervous system.

ROBOT DESIGN

4

SESSION

60 MINUTES

SOFTWARE ARCHITECTURE

KEY TOPICS

- Robot software stack
- Middleware choice
- Concurrency
- Modularity

TEACHING FOCUS

Architect robot software.

KEY TAKEAWAY Software coordinates the whole robot.

5

SESSION

60 MINUTES

INTEGRATION & TESTING

KEY TOPICS

- System integration
- Testing and validation
- Failure modes
- Iteration

TEACHING FOCUS

Assemble and verify the robot.

KEY TAKEAWAY Integration reveals the real design.

ROBOT DESIGN

6

SESSION

60 MINUTES

ROBOT DESIGN & REVIEW

KEY TOPICS

- Design review and demo
- Course recap and assessment
- Toward capstone (AI.450)
- Portfolio

TEACHING FOCUS

Present a complete robot design.

KEY TAKEAWAY A robot is judged as a whole system.